NOVA IMS
INFORMATION MANAGEMENT SCHOOL

Business Intelligence I

Fall Semester 2026/2027

Project Assignment Handout {version 1.1 13-10-2026}

This handout details the rules for the mandatory practical project for Business Intelligence I, to be developed and completed during the academic calendar of the BI I class.

Project Summary

The project for this class aims to reinforce both the conceptual and practical knowledge to be acquired from your BI course. As such, you are required, as part of a group, to design, implement and explain (in a report) a **fully working Data Warehouse solution**.

Note that a Data Warehouse is only one of the major components of a complete Business Intelligence solution; should you go onto the follow-up class next semester (BI II), you'll get the chance to build the remaining components (analysis, reporting and dashboarding layers) and finish your BI solution.

In this practical project, each group is expected to develop a proper Data Warehouse (using Microsoft Fabric Data Warehouse) and then implement a varied set of Extraction, Transformation and Loading processes (using Microsoft Fabric Data Engineering tools) to "clean" and "shape" the source data from transactional (OLTP) operational systems and load the project's data warehouse with properly structured data. Your project report will explain and detail the work that your group carried out.

Group Rules

The project should be done in a group of between one (1) to four (4) students; we consider this the ideal size for group work. Groups larger than four will NOT have project graded.

You are allowed to have group members from different BI I classes, including a mix of students from Daytime and Nighttime classes.

Important Project (partial work) Delivery Dates

First Part: **midnight 1st of November 2026**
Collection of the Source Data (loaded into Fabric Lakehouse, sampled to 20k – 30k if necessary) + description and analysis of Organization's business context + listing and explanation of project's chosen "Business Needs + Questions"

Second Part: **midnight 29th of November 2026**
Design and implementation (in Fabric, using SQL script) of the project's Data Warehouse – only the design and implementation (construction) of the DW, it is delivered EMPTY of data at this stage (also explained in Report)

Third Part: **midnight of the last Saturday before 1st call Exam**
Project Report (complete version), the Staging and DW databases (full of final, cleaned data) and all of the ETL work

Project Starting Point – The "Business" Problem and the Source Data

We strongly encourage each group to choose a real-life business/organizational problem that can be used as the foundation of this practical project. In most real-life cases, BI projects usually involve large sets of source data, from where you'll be extracting the relevant data for your project. Typically, this source data is obtained from a relational (transactional or operational) database, but this is not required for your project – most of the projects we see start from "flat" (txt or csv) files obtained from online datasets.

We expect that your source data will contain, at the very minimum, the following features (some of these will become clear as the course progresses, and you become more familiar with business intelligence terminology):

i. Some sort of transactional records that can represent quantifiable facts, to be used in your future data warehouse; these need not be extensive in volume but should represent a few thousand records of business transactions ("hard limit" of a minimum of 10k fact rows, but ideal is between 20k to 30k), also requiring some sort of "cleaning" or "normalizing" actions during the ETL process.

ii. Enough attributes (characteristics) so that you can extract at least five dimensions from the source data, according to the dimensions criteria for the project.

iii. If your source data comes from a relational database, you will have to extract the data from whatever tables you deem necessary and save this data into "flat" files (delimited text data files), so they can be loaded into Microsoft Fabric (cloud lake).

iv. The quantifiable data that will make up the facts of your data warehousing solution should represent several years of transactional history (at minimum, more than one year), so that analysis can be done across multiple years.

v. The characteristics that you'll be setting up as dimensions of your DW model should provide you with the ability to set up different hierarchies in the different dimensions ("*what is an hierarchy?*" will be explained and demonstrated in class).

vi. In order to alleviate situations where students have NOT been able to find datasets, your teachers will eventually outline some alternative open data datasets that are publicly available, and which are proven to be usable for BI projects. However, you will still be required to "imagine" and develop a problem story that fits this data, to serve as the "original data problem" that your BI solution will address.

What to do if your source data does not meet the minimum criteria above? Does "source data" quality matter?

Yes, it does... but not exactly how you may think. Essentially, what we have seen across many years of BI projects is that picking the "right" or the "wrong" source dataset can make for a much easier or a much more difficult project.

In order to try to level out this initial difficulty, we will be evaluating your source data as part of your first delivery, and giving it a "readiness score": something like 'low', 'medium' or 'high', where 'high readiness' means that the source data is already perfect for a Data Warehouse, whereas 'low readiness' means the source data is a much tougher proposition or may be incomplete in terms of data required.

However, this score DOES NOT directly affect your final project grade! It only serves to tell you how much additional and/or complementary development you must do within your BI project, in order to meet our expectations.



For example, if you receive a 'high readiness' score, you are already in an easy starting position with your source data... so we expect that you will need to 'stretch yourself' and develop additional and interesting dimension attributes, additional primitive measures, more hierarchies, many interesting and varied ETL transformations, etc. in order to demonstrate your prowess in the BI skills you will be acquiring. In other words, if you are starting from a very easy and comfortable position in terms of source data, we want to actively encourage you to use the opportunity and still develop an interesting and original project (not just do bare minimum to meet formal requirements, since this may be very straightforward to achieve and not teach you anything).

However, if you have picked a "low readiness" source dataset, you may still need to do a lot of additional work on this data in order to meet our minimum requirements (5 dimensions, 2 primitive measures, etc.). If that is the case, by documenting this additional pre-project work you are already fulfilling some of the BI skills that your delivered final solution should demonstrate... even if it ends up only meeting the minimum formal requirements.

In summary, your **source dataset** may be classified as:

- **High Readiness**, if it has all or a large mix of the following:
    - Already has 5 or more dimensions of analysis, all of them containing several rich and interesting descriptive attributes that can be directly loaded into the DW.
    - Has several (more than the minimum of 2) primitive measures, as quantifiable fields directly related to the business process and most (or all) of them are additive in nature.
    - The dimension attributes already contain several groups of descriptive attributes, enabling very easy and intuitive development of hierarchies (for example, already have the fields of product, product type and product line).
    - Most, or even all, of the fields are already in clean and appropriate format and data types, so that very little ETL transformations are actually required.
- **Low Readiness**, if it shows the following characteristics:
    - Does not have the minimum number of 5 dimensions of analysis, which means that additional dimensions would have be 'injected' into the dataset, perhaps from artificial / synthetic data.
    - Does not have at least 2 primitive measures, or the measures it holds are going to require quite a bit of ETL work in order to be extracted (for example, they may be in unstructured, free-text format); again, it will require difficult, additional work in order to develop the additional measures.
    - The dimension attributes are very sparse (very few of them or even none), not completely appropriate for analytical purposes (for example, phone numbers) or difficult to deal with.
    - The format and data types of the fields that are present in the source dataset are not ideal, as they may be not appropriate for the project (unstructured text, for example) or quite in unusual and challenging formats and data types (for example, JSON, XML, non-compliant database, etc.).

Obviously, a source dataset that is towards the middle ground between these two extremes will be deemed as being of "**Medium Readiness**".

After the first Intermediate delivery, the teacher grading your Project will provide you with specific and clear feedback on where your source dataset fits, which will allow you to better understand how much work you need to do in order to meet expected Project requirements (in particular, in terms of DW design and ETL work).



Expected Deliverables (Functional Requirements)

Your group must build and deliver a Data Warehouse that meets the following functional requirements (FR):

FR #1. A Data Warehouse, using a dimensional model (Star Schema is mandatory).

FR #2. The Data Warehouse must feature:

a. At least 5 dimensions, one of which must be a Date dimension.

b. At least 5 hierarchies (average depth of three levels when all considered).

c. At least one Fact table, featuring at least 2 distinct primitive measures.

d. At least one Fact table with a minimum of 5 thousand rows (the only exception will be for real-life projects, and subject to teacher approval).

FR #3. You are expected to employ as source data:

a. A varied set of source data, to come from transactional databases, different types of flat files (text, comma-separated files, excel, etc.), internet websites or some other "usable" sources.

b. Your Source Data must not have any individual dataset (dimension or facts) larger than 30 thousand rows; ideally, it should be between 20k to 30k (teachers will help sample down).

c. This source data must be able to be "converted" to delimited text-files (typically, CSV but Excel also works) so as to be ingested into Microsoft Fabric data lake.

FR #4. You are required to develop and use a Staging Area:

a. The staging area should be developed as a different database than the Data Warehouse.

b. The staging area database may employ tables and columns that are not part of the final DW model, and are used to make the ETL processes easier and more efficient to execute, or to check data quality.

FR #5. A set of ETL (Extract, Transform and Load) processes:

a. All ETL processes are to be developed using Microsoft Fabric Data Engineering functionality and should follow best practices seen in Labs.

b. There should be two set of independent (different) ETL processes: one process to load Staging Area, and another to load Data Warehouse.

c. As much as possible, the ETL processes should employ a varied set of techniques and transformations, reflecting the different aspects learned in the laboratory sessions.

d. The submitted ETL transformations must be self-contained inside Microsoft Fabric, not requiring any additional data or tools to run (no webservices, no calls to APIs for GenAI/LLMs, etc.

FR #6. In addition to these deliverables, you are expected to provide a complete and detailed project report that clearly describes all the work the group carried out.

IMPORTANT NOTE: Please do NOT USE as your source data a database, dataset or some other type of data that is usually employed in training / teaching / demonstration settings, as these typically have Data Warehousing solutions already available. For example, do not use AdventureWorks, Northwind, WorldWideImporters, Chinook, Sakila, etc. IN ALL CASES, IF YOU ARE DOUBTFUL ABOUT THE LEGALITY OF YOUR DATASET, ASK YOUR LAB TEACHER WELL AHEAD OF DELIVERY DEADLINE, PLEASE!

Grading Criteria for Project Delivery

GC #1. Report – the report is worth **3 points** (out of 20) of the final project grade, in accordance with the following criteria:

1.1. Structure: *is the report developed with an appropriate and proper structure, reflecting the recommendations from this Handout?*

1.2. Content: *is the report content clear and objective? Does it meet with expected guidelines for a professional BI project report? Is it efficient (not too much, nor too little) and is it effective (does it clearly explain the work done by the Group)?*

1.3. Analysis of project work: *did the group outline the methodological choices made throughout the project? Did they describe problems encountered, as well as the solutions developed to overcome those problems? Did they justify the dimensional model adopted for the Data Warehouse solution, in view of the organization's original problem?*

1.4. Conclusion: *are there concluding points, as well as a critical assessment of the project?*

GC #2. Data Warehouse – the design, implementation and use of the data warehouse is worth **8 points** (out of 20), as detailed below:

2.1. Source Data: *did the group make available the original source data that was used? Is it in "usable" format? Is it documented in the report? Are all the databases, files and/or connections also available and documented?*

2.2. Dimensional Model: *Is the dimensional model used for the Data Warehouse correct and properly developed (is the Star Schema correctly designed and implemented? Is it adjusted for the Group's specific Business Needs)?*

2.3. Staging Area: *Is the group employing a Staging Area database? Is it properly setup so as to make the ETL processes easier and more effective? Is the use of the Staging Area documented and justified, in the report?*

2.4. DW Fact table: *is the Data Warehouse's Fact table correctly designed and developed? Are the measures adequate, properly designed and configured? Are the relationships with dimensions properly configured/verifiable?*

2.5. DW Dimensions: *are the Data Warehouse's Dimension tables correctly designed and developed? Are the attributes properly designed and configured? Do the dimensions reflect the necessary attributes to enable the required hierarchies to be employed at a later stage?*

GC #3. Extraction, Transformation and Loading (ETL) Processes – the ETL processes of the project are worth **8 points** (out of 20), as detailed below:

3.1. Documentation: *Is the ETL adequately documented and explained in Report? Preferably, in a visually rich and descriptive manner, making clear the ETL approach taken by the group.*

3.2. General ETL Organization:

- *Has the workspace been delivered after a successful execution of the entire ETL process, with all STG, DW and Data Quality data fully loaded into the tables?*
- *Are the general ETL artefacts (pipelines, activities, dataflows, etc.) in Fabric properly organized and correctly named (as per Labs best practices)?*
- *Has Group met requirement in terms of mandatory use of BOTH Copy Data and Dataflows: at least ONE (1) of each MUST BE USED anywhere in the ETL.*



- *Has the Group organized their ETL into separate and properly segmented Fabric Pipelines, using distinct pipelines for different ETL objectives (i.e. STG, DW, etc.)?*

3.3. Data Quality:

- *Is the Group correctly employing Data Quality Validation checks within their ETL, fulfilling, AS A MINIMUM, the FOUR (4) Data Quality checks taught in Labs?*
- *Is the Group's ETL correctly registering and storing Data Quality validation checks into a LOG table, as specified in Labs and kept in the project's STG AREA?*

3.4. Data Transformations and Cleansing:

- *Is the project correctly employing a varied, non-trivial and technically-correct range of data transformations (e.g. cleansing) through the use of Dataflows?*
- *Are the transformations described above useful and appropriate to the project's needs? Do they make sense in relation to source data and DW design / tables?*

3.5. ETL Execution:

- *Is the STG's ETL design and implementation correct? Executes both preparatory and actual ETL tasks? Run in effective and efficient manner? Covers all STG needs?*
- *Is the DW's ETL design and implementation correct? Correctly executes preparatory tasks, SKs, lookups and complete transfer of the STG data into DW?*

GC #4. Extra ETL (Original work) – worth **1 point** (out of 20), based on 'degree of readiness / default difficulty' of the Group's chosen source dataset (see also section on page 2 about Source Data):

- Extra (original / creative) ETL development # 1
- Extra (original / creative) ETL development # 2
- Extra (original / creative) ETL development # 3
- Extra (original / creative) ETL development # 4 (max of 4; more ignored)

Remember: if your source dataset is already "low readiness", the work you do to fix it and still meet minimum requirements will, most likely, fulfill many or even all of these extra ETL developments; if, on the other hand, your source dataset is deemed to be 'high readiness', you will need to develop ALL of these additional developments in order to qualify for full project grade.

Report Structure

The structure of the project report should follow, as much as possible, the following chapter structure. Please note that the names of the chapters do NOT need to be exactly (verbatim) as per examples below, but you MUST cover the discussion points presented here; as such, we recommend that you stick to this structure as much as possible.

Chapter #1: Presentation of business / organization / problem scenario

- Presentation of the organization, providing readers (teachers) with a general context of what organization is being used, what it does and why it is a suitable candidate for the Group's BI project.
- Explanation of the informational problem that this organization is facing; in other words, what (general) business problem is the BI project solving?
- Identification of Business Needs; a more detailed – and very objective – listing of the specific business questions that are going to be driving the development of the Group's dimensional model (star schema in the DW). Please list the BN's but also provide a brief and objective explanation of each one.
- Typically, we expect this section to take around 3 to 5 "normal" pages of your Report, but varies considerably, depending on presentation choices.

Chapter #2: Original Data Sources

- Description of the structure and data in the organization's source data, as well as of any additional supporting flat files developed by Group.
- It is far more important to explain, in general terms, the source data than to simply provide a list of the original data fields.
- Typically, we expect this section to take around 1 to 2 "normal" pages of your Report, but varies considerably, depending on presentation choices (any detailed lists must go to the back of the Report, as Appendices).

Chapter #3: Staging Area

- Description of the development of the Staging Area and why it is used.
- Typically, we expect this section to take around 1 to 2 "normal" pages of your Report, but varies considerably, depending on presentation choices.

Chapter #4: Data Warehouse

- Presentation and description of the DW developed, including the methodology employed in its design, as well as how it serves to meet the business needs; must include a clear diagram of the DW, of course!
- More important than repeating "useless" (will get you zero grade points) Data Warehousing or "dimensional modelling" theory, is that you provide a clear explanation of how your choice of facts and dimensions make sense for your organization, and its informational problem.
- Typically, we expect this section to take around 2 to 6 "normal" pages of your Report, but varies considerably, depending on presentation choices.



Chapter #5: ETL Processes

- Description of the development of the ETL processes, including problems and solutions found.
- This is one section of your Report where "less is more"; it is quite important that you are exhaustive and cover the entire ETL process, but do not simply show, in repeated fashion, routine and common ETL tasks alongside trivial and inconsequential comments…
- Instead, try to show, using printscreens, diagrams and other visual artefacts, how your ETL design works, why you've designed it this way and where the "complex" (or problematic) parts are present and how they work. In other words, don't just show – please explain WHY and HOW.
- Typically, we expect this section to be the largest in the entire Report (anything from 4 or 5 pages, all the way up to 10 or more); however, the best way to expose the ETL work is by using visual imagery, to which you will add strategic and carefully considered commentary, explaining in objective but brief manner how and why each part/component of your ETL pipelines works.

Chapter #6: Listing of Extra (original) ETL Work

- Additional and original (innovative) ETL developments that were not demonstrated in the Lab classes; may have been spoken about in Labs, but if not directly demonstrated, these techniques will count as "extra work".
- This "original" work MUST be detailed in a separate section (make sure to title it appropriately), so the teachers have no chance of missing it when evaluating the Report; if we miss it, you get zero points for this work.

Chapter #7: Critical Review / Lessons Learned

- A short summary of what lessons and experience the Group, as a whole, took from this Project; Group should describe main difficulties along with positive lessons learned from having done this project.
- MOST IMPORTANT: this is separate from the Conclusion, as it covers what the Group students learned from doing this project… What "life lessons" about BI are you going to take with you, having done this project?

Chapter #8: Conclusions

- A short summary of the business value that has resulted from the development of this BI project; Group should describe, from the perspective of their Project organization, the business (additional) value that is now available to the firm, from having successfully built the Group's new BI platform.

Project Delivery Methodology

Your Project will be evaluated across **three (3) partial but cumulative deliveries**.

This both encourages each group to start working on the BI project as soon as possible, but also ensures that you have some means of responding to feedback and (partially) correct and realign your BI journey, in future deliveries.

In each delivery, you MUST BOTH submit part of your Project Report (in pdf format) in the respective Moodle submission section, as well as make available (by sharing with all Lab teachers) your Project Fabric Workspace, containing the work to be delivered:

- **Delivery #1 – due at 23h59 of 1st of November of 2026:**
  - In this delivery, you must submit all the work referred to in GC #2.1 of the Grading Criteria (in essence, the Source data).
  - You must also submit your Project Report, which should – at this stage – contain Chapters #1 and #2 referred to in the Report Structure.
- **Delivery #2 – due at 23h59 of 29th of November of 2026:**
  - In this delivery, you must submit all the work referred to in section GC #2 of the Grading Criteria (all of the DW work) EXCEPT the Staging Area (GC #2.3); your DW is to be delivered EMPTY, at this stage.
  - You must also submit your Project Report, which should – at this stage – add Chapter #4 referred to in the Report Structure, to previous delivery.
- **Delivery #3 - due at 23h59 of last Saturday before the 1st Call Exam for BI I:**
  - In this delivery, you must submit all the remaining work referred to in the Grading Criteria; in other words, please complete and deliver anything that has not yet been evaluated in previous deliveries.
  - You must also submit your Project Report, which should now be complete and in its final presentation format.

Please note that there is no REGRADING of Intermediate deliveries; in other words, the criteria that counts for Intermediate evaluation are only graded once; you can (and should), of course, change and improve your project after Intermediate delivery feedback, but those parts are NOT REGRADED.

Applicable for ALL deliveries, in cumulative fashion:

- **Failure to deliver on time will incur a 0.5 point penalty for each late day** (for example, 4 late days will accrue a 2.0 point penalty)**, across any of the deliveries (in cumulative fashion).**
- **Failure to comply with Moodle and/or Fabric delivery instructions** (missing student identification, no proper naming of objects, duplicated or unclear files, improper folder configuration, etc.) will meet with a **0.5 point penalty** at each delivery where this failure is present and detected.



Questions and Clarifications

Should it be necessary, we will provide further clarifications and answers to questions from students, updating the respective Moodle project forum as appropriate.

Good luck with your project!